EquiCut: A Web Tool for Determining Empirical Cut-Off Points Using an Entropy Measure

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Abstract

(Background): The dichotomization of continuous or ordinal scales is a common practice in psychometric and survey research, but the justification for choosing a specific cut-off point is often weak, especially in the absence of an external criterion. **(Tool Presentation):** We introduce EquiCut, a free, open-source, and user-friendly web tool designed to address this challenge. **(Methodology):** EquiCut implements an information-theoretic method based on Shannon's entropy. The entropy value for a score distribution is first computed, and an equivalent equitability index is obtained by normalizing it by the number of categories. The optimal cut-off is identified by finding the dichotomized distribution whose equitability index best matches that of the original multi-category distribution, thus preserving its structural properties. **(Features):** The tool requires no programming knowledge or software installation. Users simply upload a frequency data file, and the tool provides a comprehensive report. **(Availability):** The web tool and its full Python source code are freely available.

Keywords: dichotomization, cut-off point, entropy, equitability index, software, web tool, psychometrics, Kessler K6

1 Introduction: The Challenge of Cut-Off Point Selection

In psychometric and health survey research, it is common practice to dichotomize continuous and ordinal scales to create clinically or practically meaningful groups (e.g., "high distress" vs. "low distress"). However, the rationale for selecting a specific cut-off point is often poorly documented or relies on non-empirical methods, such as a median split or the adoption of thresholds from previous studies. It has also been well established that dichotomization inevitably results in a loss of information and a reduced ability to detect effects (MacCallum et al., 2002; Streiner, 2002). An ideal approach should therefore not only justify the choice of threshold empirically, but also seek to minimize this informational loss.

While data-driven methods exist, the most common approach—Receiver Operating Characteristic (ROC) curve analysis—requires a "gold standard" external criterion, such as a clinical diagnosis. In many research contexts, particularly large-scale population health surveys, such criteria are unavailable. At a population level, any empirically derived cut-off point is fundamentally a function of the score's own distributional properties. An ideal method should therefore be able to leverage these properties directly to provide a robust, data-driven rationale for dichotomization.

To address this methodological gap, we developed EquiCut, a web-based tool that implements an empirical method based on Shannon's entropy (Shannon, 1948; Stewart, 2012). The purpose of this note is to introduce the tool and provide a practical guide for its use, empowering researchers to apply this information-theoretic approach without needing programming expertise.

2 Method and Implementation

2.1 The Entropy-Based Method

The statistical method implemented in EquiCut originates from information theory. The core of the method is the calculation of Shannon's entropy (H), which measures the amount of uncertainty in a distribution. To make this measure comparable across distributions with different numbers of categories (e.g., a 25-category scale vs. a 2-category dichotomization), the entropy value is normalized.

This normalization is achieved by dividing the calculated Shannon entropy (H) by the log of the number of categories ($\ln S$, where S is the number of categories). The result is the equitability index (E), a value ranging from 0 to 1 that represents evenness, or the degree of dispersion, of the distribution:

$$E = \frac{H}{\ln S} \tag{1}$$

This formulation is available in statistical software such as PAST (Hammer et al., 2001). The tool first computes the overall equitability index for the original multi-category score distribution. It then systematically calculates the equitability index for every possible dichotomized (two-category) version of the data. The optimal cut-off point is identified as the one where the absolute difference between the overall equitability index and the dichotomized equitability index is minimized. This process ensures the selected cut-off best preserves the structural shape of the original data. For a full theoretical description and validation of the method, which was validated using the K6 scale for psychological distress (Kessler et al., 2002), readers are referred to Marleau & Garvie (2026).

2.2 The EquiCut Algorithm

The logic of the EquiCut tool is presented in Algorithm 1. To ensure reproducibility, the algorithm identifies the cut-off c that minimizes the information divergence between the original scale and the simplified binary classification.

Algorithm 1 Optimal Cut-Off Selection via Equitability Matching

Require: Vector of frequencies F for S categories (scores 0 to $S-1$).

Step 1: Calculate Global Equitability (E_{global})

$N \leftarrow \sum_{i=0}^{S-1} F_i$ ▷ Total sample size

$H \leftarrow -\sum_{i=0}^{S-1} \frac{F_i}{N} \times \ln \frac{F_i}{N}$ ▷ Shannon Entropy

$E_{\text{global}} \leftarrow H / \ln S$

Step 2: Iterative Dichotomization

for each cut-off c from 0 to $S-2$ **do**

  $\text{Count}_A \leftarrow \sum_{i=0}^{c} F_i$ ▷ Group A: Low scores

  $\text{Count}_B \leftarrow \sum_{i=c+1}^{S-1} F_i$ ▷ Group B: High scores

  Calculate H_{cutoff} for the 2-group distribution $\{\text{Count}_A, \text{Count}_B\}$

  $E_{\text{cutoff}} \leftarrow H_{\text{cutoff}} / \ln 2$

  $\Delta_c \leftarrow |E_{\text{global}} - E_{\text{cutoff}}|$

end for

Ensure: Cut-off point $c/(c+1)$ that minimizes Δ_c

2.3 Software Implementation

EquiCut is implemented in Python (v3.9+). The interactive web interface is built using the Streamlit library, and data handling is managed by Pandas. The application is hosted on the Streamlit Community Cloud, making it accessible from any modern web browser without requiring any software installation or user registration.

3 Example Application: A Step-by-Step Walkthrough

Using the tool involves a simple, four-step process. We demonstrate this workflow using the K6 psychological distress scale data from the 2009–2010 Canadian Community Health Survey, as analyzed in Marleau & Garvie (2026).

Step 1: Data Preparation The required input is a single-column text file (`.txt`). Each line in the file must contain the frequency count for each successive category of the scale, starting from the lowest score (e.g., score 0).

Step 2: Uploading Data and Running the Analysis The user navigates to the EquiCut web application and uploads their prepared text file using the file browser or by dragging and dropping the file. Once uploaded, the user clicks the "Run Analysis" button (Figure 1).

Step 3: Interpreting the Results The application processes the data and displays a full report (Figure 2). The report includes: (a) a summary of global results; (b) the optimal cut-off point, highlighted for clarity; and (c) a detailed table presenting the equitability index for every threshold.

Step 4: Downloading the Report A full text report can be downloaded by clicking the "Download results file" button at the bottom of the results page.

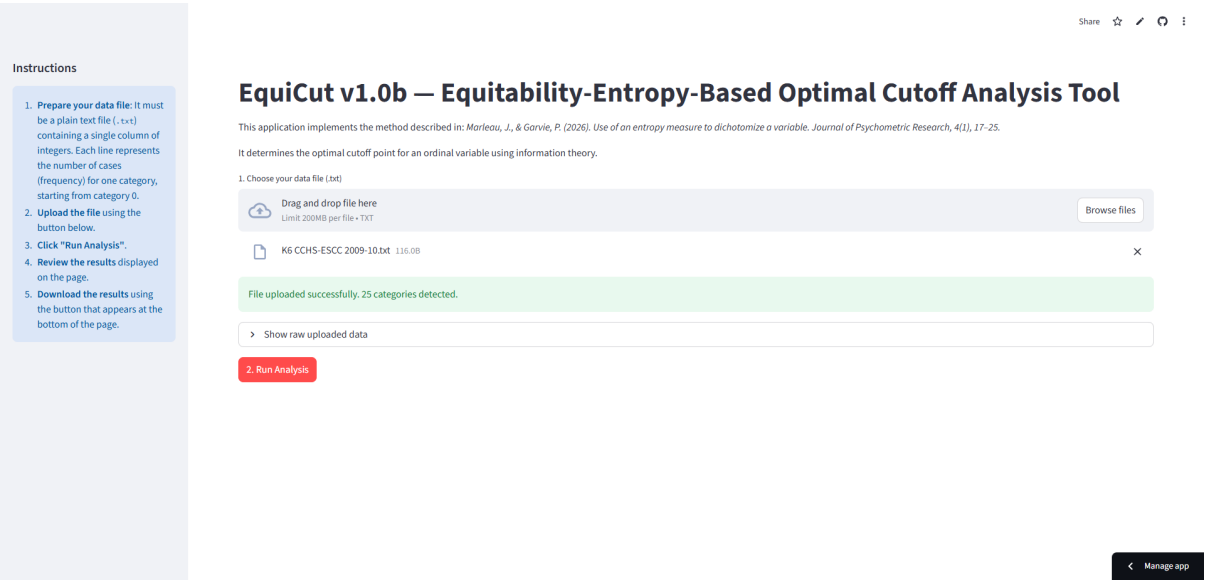
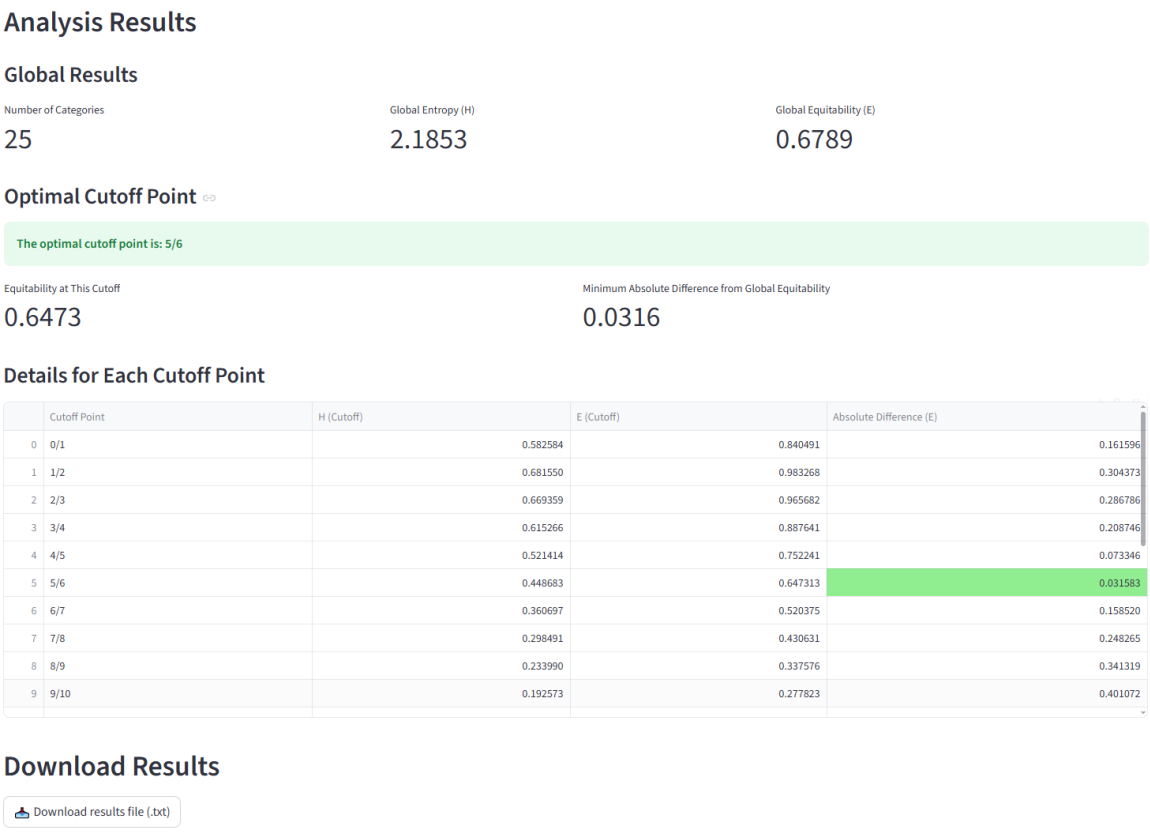


Figure 1: The EquiCut user interface showing instructions and file upload panel.



Analysis Results

Global Results

Number of Categories
25

Global Entropy (H)
2.1853

Global Equitability (E)
0.6789

Optimal Cutoff Point

The optimal cutoff point is: 5/6

Equitability at This Cutoff
0.6473

Minimum Absolute Difference from Global Equitability
0.0316

Details for Each Cutoff Point

	Cutoff Point	H (Cutoff)	E (Cutoff)	Absolute Difference (E)
0	0/1	0.582584	0.840491	0.161596
1	1/2	0.681550	0.983268	0.304373
2	2/3	0.669359	0.965682	0.286786
3	3/4	0.615266	0.887641	0.208746
4	4/5	0.521414	0.752241	0.073346
5	5/6	0.448683	0.647313	0.031583
6	6/7	0.360697	0.520375	0.158520
7	7/8	0.298491	0.430631	0.248265
8	8/9	0.233990	0.337576	0.341319
9	9/10	0.192573	0.277823	0.401072

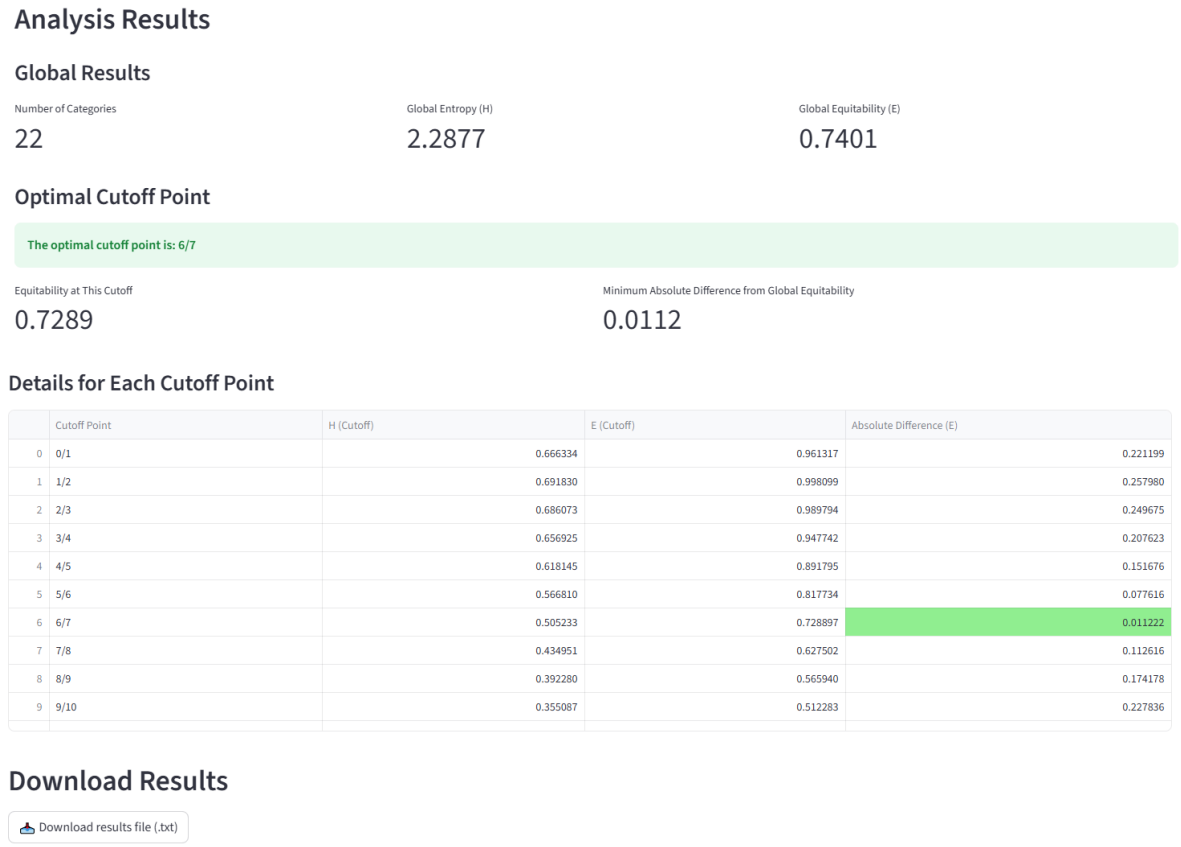
Download Results

Download results file (.txt)

Figure 2: The results page generated by EquiCut for the K6 displaying the optimal cut-off (5/6).

3.1 Generalizability of EquiCut

To further illustrate the generalizability of EquiCut beyond the K6, a second example is provided using the Generalized Anxiety Disorder scale (GAD-7; Spitzer et al. 2006), a widely used instrument for assessing anxiety symptoms. The data were collected from a survey of 10,513 adults in Québec in February 2021, during the COVID-19 pandemic. GAD-7 scores range from 0 to 21, yielding 22 categories. Applying EquiCut to this distribution identifies 6/7 as the optimal cut-off point (Figure 3).



Analysis Results

Global Results

Number of Categories: 22

Global Entropy (H): 2.2877

Global Equitability (E): 0.7401

Optimal Cutoff Point

The optimal cutoff point is: 6/7

Equitability at This Cutoff: 0.7289

Minimum Absolute Difference from Global Equitability: 0.0112

Details for Each Cutoff Point

	Cutoff Point	H (Cutoff)	E (Cutoff)	Absolute Difference (E)
0	0/1	0.666334	0.961317	0.221199
1	1/2	0.691830	0.998099	0.257980
2	2/3	0.686073	0.989794	0.249675
3	3/4	0.656925	0.947742	0.207623
4	4/5	0.618145	0.891795	0.151676
5	5/6	0.566810	0.817734	0.077616
6	6/7	0.505233	0.728897	0.011222
7	7/8	0.434951	0.627502	0.112616
8	8/9	0.392280	0.565940	0.174178
9	9/10	0.355087	0.512283	0.227836

Download Results

Download results file (.txt)

Figure 3: The results page generated by EquiCut for the GAD-7 displaying the optimal cut-off (6/7).

Both datasets — the K6 (CCHS 2009–2010) and the GAD-7 (EQSP 2020–21) — are available in the EquiCut GitHub repository, allowing researchers to independently verify the results and to familiarize themselves with the required input file format.

4 Guidelines for Interpretation and Expert Judgment

While EquiCut identifies the mathematical point of information equivalence, it should be viewed as a tool to identify potential candidate cut-off points. The researcher's expertise remains paramount in selecting the final threshold among the top candidates identified by the entropy measure.

4.1 The Role of Content Validity and Expert Judgment

A critical feature of the EquiCut output is the full results table, which allows researchers to inspect the performance of all possible thresholds. Methodologically, it is essential not to follow the 'optimal' recommendation blindly. In certain scenarios—particularly when a population distribution becomes more uniform (due to external stressors)—the algorithm may identify multiple local minima with near-identical equitability differences.

In such cases, the final selection must be guided by content validity. The researcher must evaluate whether the suggested cut-off point aligns with the theoretical construct being measured. For example, if a suggested cut-off is mathematically optimal but theoretically separates individuals with no symptoms from those with minimal symptoms, the researcher may instead opt for a slightly higher candidate threshold that better identifies a 'case' in the psychometric or clinical sense. EquiCut provides the empirical evidence for this choice, but the decision must remain anchored in the nature of the scale and the research objectives.

To summarize, EquiCut offers a transparent, data-driven rationale for dichotomization. However, it is designed to complement, not replace, expert clinical or psychometric judgment. The integration of information theory with content validity ensures that the chosen cut-off point is both statistically robust and theoretically sound.

5 Availability and Reproducibility

- **Tool Name:** EquiCut
- **Web Application:** https://equicut.streamlit.app/
- **Source Code:** https://github.com/marleaugarvie/EquiCut/tree/main
- **Reproducibility:** Both datasets `CCHS-ESCC 2009-10 K6.txt` and `EQSP 2020-2021 GAD-7.txt` are included in the GitHub repository to allow researchers to verify the tool's output.
- **License:** MIT License. The source code is provided under this permissive license to encourage transparency and further development.

- **Citation:** Researchers who use EquiCut are encouraged to cite this article and the primary methodological paper (Marleau & Garvie, 2026).

6 Conclusion and Future Directions

EquiCut bridges the gap between a robust information-theoretic method and practical application. It provides an empirical, transparent, and data-driven solution to the selection of a cut-off point when no external criterion is available. Future development plans include the integration of data visualization plots and a database to allow users to optionally share their findings with other researchers.

Use of Artificial Intelligence

Language editing assistance was used in some sections of this manuscript to improve clarity and readability. All scientific content, methodological decisions, and interpretations were made solely by the authors.

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